ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Right-Angled Triangles - Pythagoras & Trigonometry

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

RIGHT-ANGLED TRIANGLES - PYTHAGORAS & TRIGONOMETRY · CH4 · L10

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The Map

This lesson collapses Right-Angled Triangles - Pythagoras & Trigonometry into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Name the angle relationship

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use the polygon sum

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Use exterior angles

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Chain reasons in order

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Angle

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 32 website questions, 32 checked solutions, 3 local PDFs read.

01 Name the angle relationship

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Right-Angled Triangles - Pythagoras & Trigonometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\cos 63^\circ = \frac{24.3}{PQ}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q9

Use trigonometry

$$\cos 63^\circ = \frac{24.3}{PQ}$$

Finish: 53.5 cm.

FORMULA BANK

SOH CAH TOA and $a^2 + b^2 = c^2$. Choose from the labelled sides.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

> A right-angled triangle has hypotenuse 14.2 cm and angle 37°. Find the side opposite the angle.

YOUR SOLUTION

02 Use the polygon sum

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Right-Angled Triangles - Pythagoras & Trigonometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$EF = 12\cos 40^\circ = 9.192\ldots$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q15

Use trigonometry

$$EF = 12\cos 40^\circ = 9.192\ldots$$

Use trigonometry

$$\sin 28^\circ = \frac{EF}{EG}$$

Finish: 19.6 cm.

FORMULA BANK

SOH CAH TOA and $a^2 + b^2 = c^2$. Choose from the labelled sides.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

> A right-angled triangle has hypotenuse 14.2 cm and angle 37°. Find the side opposite the angle.

YOUR SOLUTION

03 Use exterior angles

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Right-Angled Triangles - Pythagoras & Trigonometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\cos 63^\circ = \frac{24.3}{PQ}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q9

Use trigonometry

$$\cos 63^\circ = \frac{24.3}{PQ}$$

Finish: 53.5 cm.

FORMULA BANK

SOH CAH TOA and $a^2 + b^2 = c^2$. Choose from the labelled sides.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

> A right-angled triangle has hypotenuse 14.2 cm and angle 37°. Find the side opposite the angle.

YOUR SOLUTION

04 Chain reasons in order

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Right-Angled Triangles - Pythagoras & Trigonometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$BD = 16, \qquad \angle DAB = 65^\circ$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q10

Find the gradient
This is a reasoning step: write one clear sentence, then continue the calculation.

Use trigonometry

$$BD = 16, \qquad \angle DAB = 65^\circ$$

Finish: 50.2 cm.

FORMULA BANK

SOH CAH TOA and $a^2 + b^2 = c^2$. Choose from the labelled sides.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

> A right-angled triangle has hypotenuse 14.2 cm and angle 37°. Find the side opposite the angle.

YOUR SOLUTION

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Right-Angled Triangles - Pythagoras & Trigonometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\cos 63^\circ = \frac{24.3}{PQ}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q9

Use trigonometry

$$\cos 63^\circ = \frac{24.3}{PQ}$$

Finish: 53.5 cm.

FORMULA BANK

SOH CAH TOA and $a^2 + b^2 = c^2$. Choose from the labelled sides.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

> A right-angled triangle has hypotenuse 14.2 cm and angle 37°. Find the side opposite the angle.

YOUR SOLUTION

06 Angle

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Right-Angled Triangles - Pythagoras & Trigonometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$AB^2 = 21^2 - 15^2 = 216$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Nov 2025 4WM1H Q11

Use trigonometry

$$AB^2 = 21^2 - 15^2 = 216$$

Angle

$$\cos ACB = \frac{15}{21}$$

Finish: 13.5°.

FORMULA BANK

SOH CAH TOA and $a^2 + b^2 = c^2$. Choose from the labelled sides.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

> A right-angled triangle has hypotenuse 14.2 cm and angle 37°. Find the side opposite the angle.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Name the angle relationship
"work out", "show", "hence", a familiar exam phrase	Use the polygon sum
"work out", "show", "hence", a familiar exam phrase	Use exterior angles
"work out", "show", "hence", a familiar exam phrase	Chain reasons in order
"work out", "show", "hence", a familiar exam phrase	Use trigonometry
"work out", "show", "hence", a familiar exam phrase	Angle

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.